TITLE

E-Commerce Platform: Selecting the Most Suitable Open-Source Web Framework

[NAME – Yatharthsinh Zala] [ENROLLMENT – 92400103336]

DEPARTMENT
COMPUTER ENGINEERING

ACADEMIC YEAR

2026–2027

SEMESTER

5TH

SUBJECT
OPEN SOURCE TECHNOLOGIES (01CE0526)
CLASS
EC3
BATCH

A

RUBRICS FOR COMPLEX ENGINEERING PROBLEM

Sr. No.	Assessment Criteria	Excellent	Good	Satisfactory	Poor	Marks
1	Problem Analysis & Open Source Technology Selection	3 Marks: Clearly identifies the engineering problem, analyzes requirements/constraints, and selects appropriate Open Source technologies with proper justification.	2 Marks: Good problem analysis with suitable technology selection and minor gaps.	1 Mark: Basic analysis with limited justification of technology selection.	0 Marks: Inadequate problem analysis and inappropriate technology selection.	3
2	System Design, Implementation Strategy & Technical Analysis	4 Marks: Well-defined system architecture, appropriate technology stack, database, deployment strategy, version control, and comprehensive technical analysis including scalability, reliability, and security.	3 Marks: Good system design and technical discussion with minor omissions.	1–2 Marks: Basic architecture and limited technical analysis.	0 Marks: Poor or missing system design and technical analysis.	4

Sr. No.	Assessment Criteria	Excellent	Good	Satisfactory	Poor	Marks
3	Testing, Documentation & References	3 Marks: Comprehensive testing strategy, well-organized report, IEEE references, appropriate diagrams, and clear conclusions.	2 Marks: Good documentation with minor formatting or testing gaps.	1 Mark: Basic report with limited testing or references.	0 Marks: Incomplete report with poor documentation and missing references.	3
	Total					10

Sr. No.	Complex Engineering Problem Activity / Problem Title	COs	Date	R1(4)	R2(3)	R3(4)	Marks (10)	Signature
1	E-Commerce Platform: Selecting the Most Suitable Open-Source Web Framework	CO1–CO5						

Notes / Instructions

1. CEP activity must be completed in a group of maximum 3 students.
2. Students must strictly follow the given deadlines for submission.
3. The report must be prepared in the prescribed format and submitted as a DOCX file on Google Classroom.
4. A hard copy of the report must be submitted as discussed.
5. The report should include problem analysis, design/architecture, implementation, results, and challenges faced.
6. All group members must contribute equally and be prepared for evaluation/presentation.
7. Plagiarism or copying from other groups will not be accepted.
8. Each group member must understand the complete solution and should be able to explain their contribution during evaluation.

Ms. Priyanka Mangi Subject Faculty

TITLE

E-Commerce Platform: Selecting the Most Suitable Open-Source Web Framework

Course Name and Course Code: Open Source Technologies (01CE0526)
CO's Mapped – CO1 (OSS concepts), CO2 (Version control), CO3 (Web application development), CO4 (Database technologies), CO5 (Testing, deployment & automation)
SDG's Mapped – SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), SDG 12 (Responsible Consumption and Production)

1. Aim

To analyse the requirements of an E-commerce platform and select the most suitable open-source web framework by comparing candidate frameworks against weighted engineering criteria, using appropriate knowledge and tools.

2. Problem Statement

A startup wants to build a scalable, secure, and cost-effective E-commerce platform for its university project and needs to select the most suitable open-source web framework among the available options, considering performance, scalability, security, payment-gateway integration, and development effort.

3. Problem Analysis

a) What is the engineering problem?
The problem is to evaluate multiple open-source web frameworks against the specific demands of an E-commerce platform — product catalogue scale, secure payment handling, concurrent checkout load, and development speed — and to select and justify the single most suitable framework for building the platform.

b) Why is this problem complex? ☒ Requires multiple engineering concepts
☒ Requires analysis and decision making
☒ Involves design constraints
☒ Requires use of engineering tools
☒ Requires performance/security/safety consideration

4. Knowledge and Tools Required

Engineering concepts used:
1. MVC/MVT architectural patterns and framework design philosophy
2. E-commerce domain concepts (catalogue, cart, checkout, order lifecycle)
3. Payment-gateway integration and PCI-relevant security awareness
4. Performance and load-testing concepts for concurrent checkout traffic

5. Weighted decision-matrix analysis for technology selection 6. Database design for product, order and user data **Tools/Software used:**
1. Candidate frameworks: Django (Python), Laravel (PHP), Node.js/Express (JavaScript)
2. PostgreSQL/MySQL, Git and GitHub, Docker
3. Apache JMeter (performance benchmarking), OWASP ZAP (security scanning)
4. Stripe/PayPal sandbox environment (payment-gateway testing)

5. Proposed Solution Design

Selection Methodology:

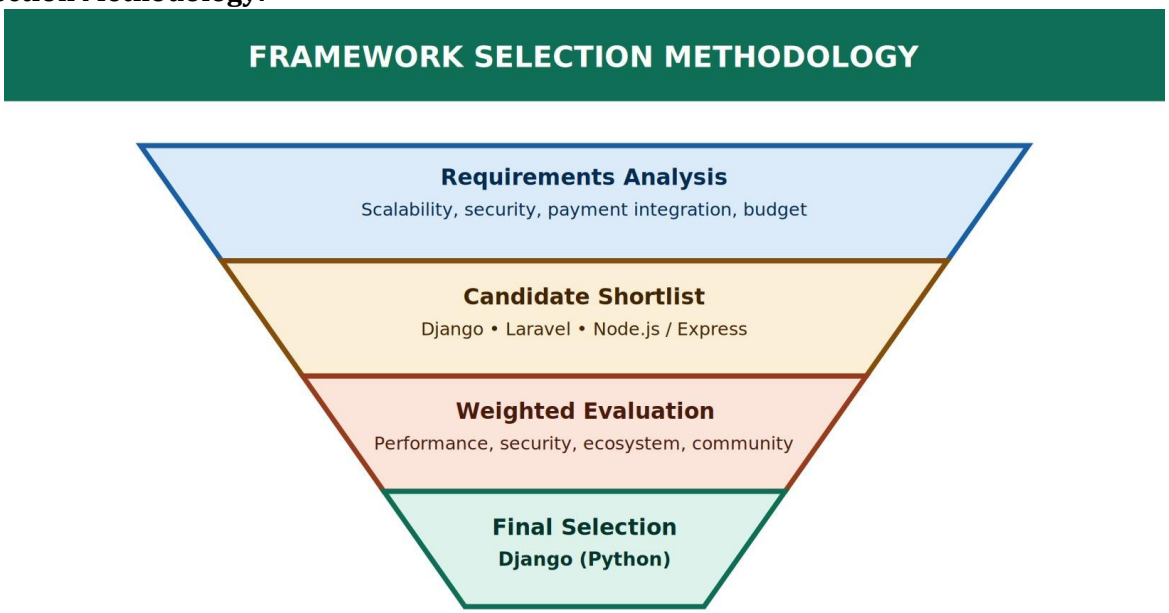


Fig. 1. Four-stage methodology used to shortlist and select the framework.

Weighted Comparison:

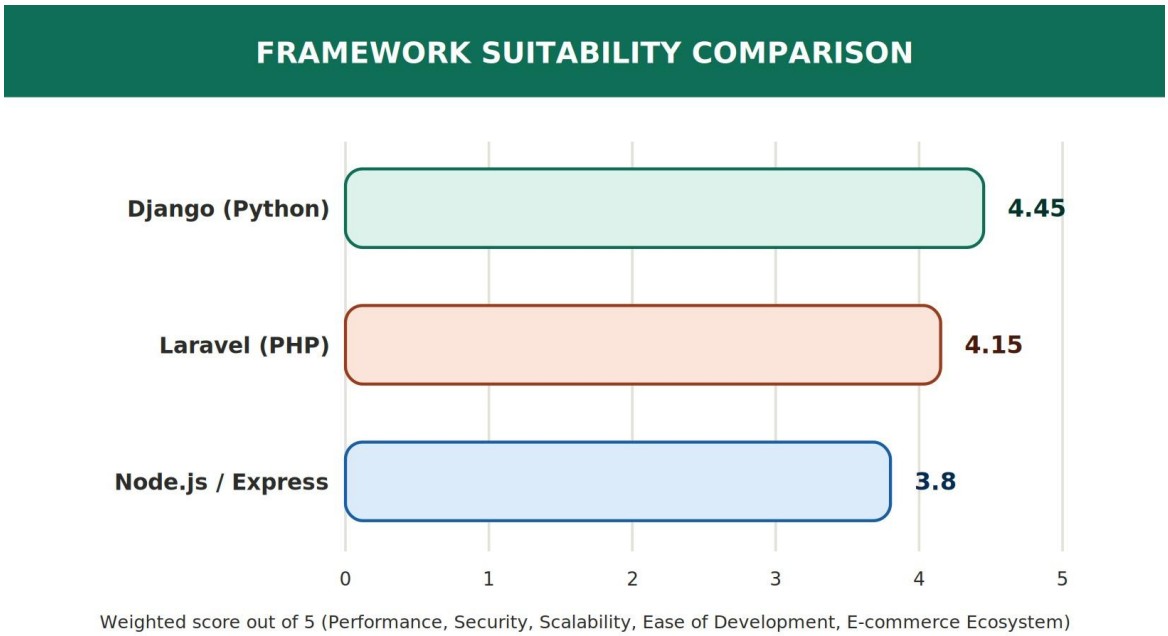


Fig. 2. Weighted suitability scores for the three candidate frameworks.

Proposed Platform Architecture (selected framework):

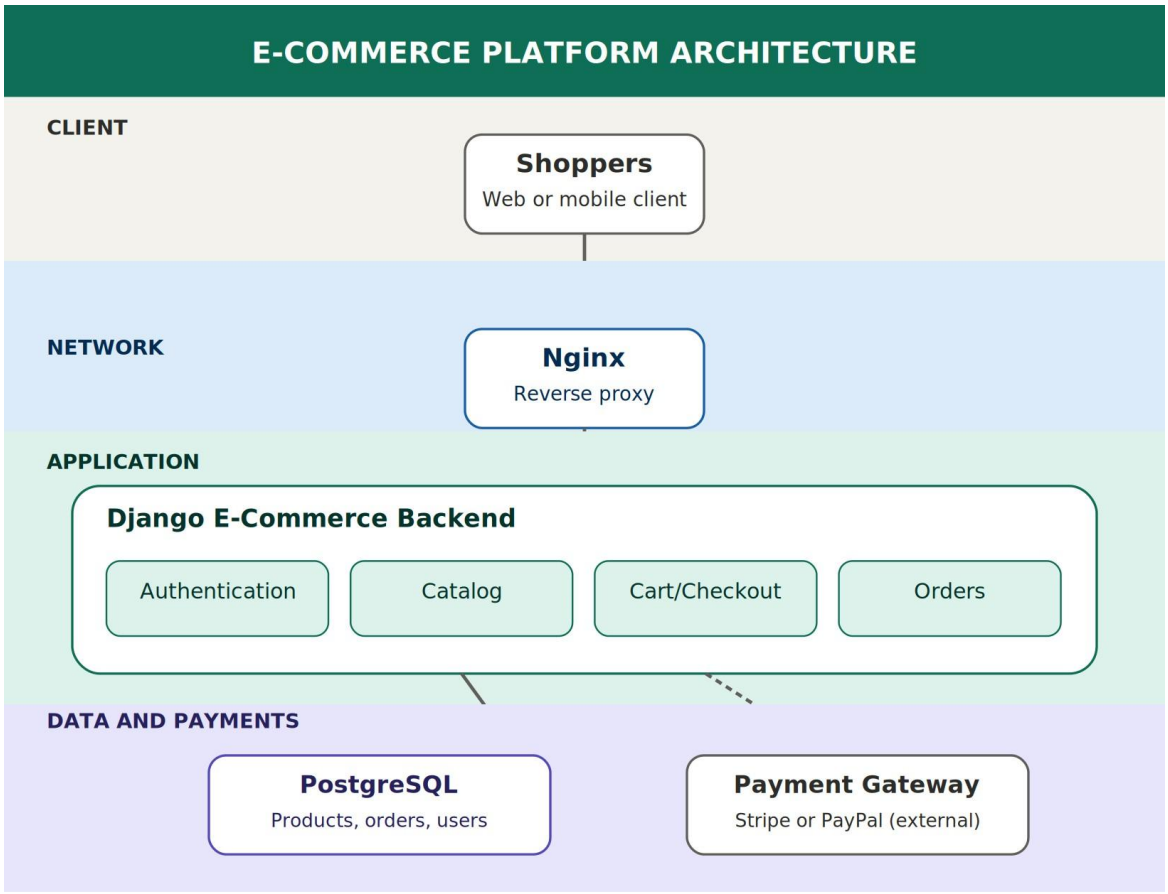


Fig. 3. E-commerce platform architecture built on the selected framework.

Brief Working:

Three open-source candidates — Django, Laravel, and Node.js/Express — were shortlisted and scored against five weighted criteria: performance, security, scalability, ease of development, and Ecommerce ecosystem maturity (existing open-source E-commerce packages built on that framework). Django scored highest overall (4.45/5), primarily due to its mature E-commerce ecosystem (Saleor, django-oscar), built-in security defaults (CSRF protection, ORM-based SQLinjection prevention), and strong ORM for relational product/order data, despite Node.js/Express scoring higher on raw performance and Laravel scoring higher on ease of development. In the proposed architecture, shoppers reach the platform over HTTPS through Nginx, which forwards requests to the Django backend's authentication, catalogue, cart/checkout, and order-management modules; catalogue and order data are persisted in PostgreSQL, while the checkout module calls an external payment gateway (Stripe or PayPal) over a secure API rather than handling card data directly, reducing PCI compliance scope.

6. Implementation / Procedure

Step 1: Define E-commerce functional requirements: catalogue browsing, cart, checkout, order history, admin panel.
Step 2: Shortlist candidate open-source frameworks: Django, Laravel, Node.js/Express.
Step 3: Define weighted evaluation criteria and assign weights based on project priorities. Step 4: Build a minimal proof-of-concept storefront (product list + cart) in each shortlisted framework.
Step 5: Benchmark each proof-of-concept under simulated concurrent load using Apache JMeter.
Step 6: Run an OWASP ZAP baseline scan against each proof-of-concept to compare default security posture.
Step 7: Score each framework against the weighted criteria and compute the final weighted total.
Step 8: Select the highest-scoring framework and build out the full platform architecture on it.
(Insert screenshots of each proof-of-concept storefront, JMeter results, and the ZAP scan report here, each numbered and captioned, e.g. Fig. 4 – JMeter load-test results.)

7. Testing and Results

Test/Parameter	Expected Result	Obtained Result
Load test – 200 concurrent users on product listing (JMeter)	Average response time within acceptable threshold	Django and Node.js remained within threshold; Laravel close to threshold
Security scan – default configuration (OWASP ZAP)	No high-severity vulnerabilities on default setup	Django flagged fewest default issues; Node.js required additional middleware to match
Development time – build cart & checkout POC	Feature complete within sprint estimate	Laravel fastest to a working POC; Django close second
E-commerce package availability	Mature, actively maintained open-source packages	Django (Saleor, django-oscar) and Laravel (Bagisto) well supported; Node.js more fragmented
Weighted overall suitability score (out of 5)	Highest score selected as final framework	Django 4.45, Laravel 4.15, Node.js/Express 3.80 — Django selected

8. Challenges Faced and Solutions

Challenge	Solution
Comparing frameworks fairly despite differing built-in features	Built an equivalent minimal proof-of-concept in each framework and measured the same criteria under the same conditions
Benchmarks were sensitive to hosting/hardware differences	Ran all load tests on identical containerised environments (Docker) to keep conditions consistent
Balancing raw performance (Node.js) against ecosystem maturity (Django)	Used a weighted decision matrix so no single criterion dominated the outcome, keeping the choice defensible
Avoiding storage of sensitive card data in the platform	Delegated payment handling to an external gateway (Stripe/PayPal) via API instead of processing cards directly

9. Conclusion

This project analysed the requirements of an E-commerce platform and selected the most suitable open-source web framework using a structured, weighted evaluation methodology rather than a subjective choice. Among Django, Laravel, and Node.js/Express, Django was selected with a weighted suitability score of 4.45 out of 5, on the strength of its mature E-commerce ecosystem, strong default security posture, and relational data handling suited to product, order, and user data, while Laravel and Node.js/Express remain credible alternatives favouring development speed and raw performance respectively. The outcome is a justified framework selection and a proposed layered platform architecture — Nginx, Django (authentication, catalogue, cart/checkout, orders), PostgreSQL, and an external payment gateway — that the university project team can build on with confidence in the underlying technology choice.

10. SDG Mapping and Justification

SDG 8 – Decent Work and Economic Growth: An open-source E-commerce platform lowers the barrier for small businesses and student entrepreneurs to sell online without expensive proprietary licensing.
SDG 9 – Industry, Innovation and Infrastructure: The project applies a structured, engineering decision-making process to build reliable digital commerce infrastructure.

SDG 12 – Responsible Consumption and Production: Selecting a mature, well-supported opensource framework reduces duplicated engineering effort and long-term technical waste compared to a poorly-fit or abandoned platform.

11. References (IEEE Style)

[1] Django Software Foundation, "Django Documentation," [Online]. Available: https://docs.djangoproject.com. [Accessed: 2026].
[2] Laravel LLC, "Laravel Documentation," [Online]. Available: https://laravel.com/docs. [Accessed: 2026].
[3] OpenJS Foundation, "Express.js Documentation," [Online]. Available: https://expressjs.com. [Accessed: 2026].
[4] OWASP Foundation, "OWASP ZAP," [Online]. Available: https://www.zaproxy.org/docs/. [Accessed: 2026].